

Exercise 10 - Bayesian Non-parametrics and Bayesian Bootstrapping

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Setup

```
set.seed(12242987)
library(knitr)
```

```
## Warning: package 'knitr' was built under R version 4.4.2
```

```
library(DPpackage)
```

```
##
```

```
## DPpackage 1.1-7
```

```
##
```

```
## Copyright (C) 2006 - 2012, Alejandro Jara
```

```
## Department of Statistics
```

```
## P.U. Catolica de Chile
```

```
##
```

```
## Support provided by Fondecyt
```

```
## 11100144 grant.
```

```
##
```

Simulate data

```

n <- 500
x <- runif(n, 0, 1)
y <- sapply(x, function(xi) {
  if (runif(1) < exp(-2 * xi)) {
    rnorm(1, mean = 0, sd = 0.1)
  } else {
    rnorm(1, mean = xi^2, sd = 0.2)
  }
})
data <- data.frame(x = x, y = y)
kable(head(data), caption = "Generated data")

```

Table 1: Generated data

x	y
0.9742491	0.8584634
0.9671887	-0.0532718
0.3986076	0.0569975
0.1384320	-0.0044624
0.6406802	0.5960664
0.1425395	0.0195705

Fit the DPdensity model

```

# Define hyperparameters
prior <- list(
  a0 = 10,
  b0 = 1,
  nu1 = 4,
  nu2 = 4,
  m2 = c(0, 0), # mean vector for the prior distribution
  s2 = diag(2), # covariance matrix for the prior
  psiinv2 = solve(diag(2)), # inverse covariance matrix
  tau1 = 6.01,
  tau2 = 3.01
)

# MCMC parameters
mcmc <- list(
  nburn = 5000, # burn-in period
  nsave = 5000, # number of samples to save
  nskip = 4, # thinning interval
  ndisplay = 100 # display every 100 iterations
)
fit <- DPdensity(
  y = y,
  prior = prior,
  mcmc = mcmc,
  state = NULL,

```

```
status = TRUE
)
```

```
## Warning in matrix(prior$psiinv2, nvar, nvar): data length differs from size of
## matrix: [4 != 1 x 1]
```

```
##
## MCMC scan 100 of 5000 (CPU time: 4.797 s)
## MCMC scan 200 of 5000 (CPU time: 5.609 s)
## MCMC scan 300 of 5000 (CPU time: 6.969 s)
## MCMC scan 400 of 5000 (CPU time: 7.719 s)
## MCMC scan 500 of 5000 (CPU time: 8.844 s)
## MCMC scan 600 of 5000 (CPU time: 10.484 s)
## MCMC scan 700 of 5000 (CPU time: 11.797 s)
## MCMC scan 800 of 5000 (CPU time: 13.094 s)
## MCMC scan 900 of 5000 (CPU time: 13.844 s)
## MCMC scan 1000 of 5000 (CPU time: 14.469 s)
## MCMC scan 1100 of 5000 (CPU time: 15.219 s)
## MCMC scan 1200 of 5000 (CPU time: 16.219 s)
## MCMC scan 1300 of 5000 (CPU time: 17.328 s)
## MCMC scan 1400 of 5000 (CPU time: 18.172 s)
## MCMC scan 1500 of 5000 (CPU time: 19.625 s)
## MCMC scan 1600 of 5000 (CPU time: 20.609 s)
## MCMC scan 1700 of 5000 (CPU time: 21.531 s)
## MCMC scan 1800 of 5000 (CPU time: 22.531 s)
## MCMC scan 1900 of 5000 (CPU time: 23.609 s)
## MCMC scan 2000 of 5000 (CPU time: 24.891 s)
## MCMC scan 2100 of 5000 (CPU time: 25.953 s)
## MCMC scan 2200 of 5000 (CPU time: 27.188 s)
## MCMC scan 2300 of 5000 (CPU time: 28.562 s)
## MCMC scan 2400 of 5000 (CPU time: 29.703 s)
## MCMC scan 2500 of 5000 (CPU time: 31.078 s)
## MCMC scan 2600 of 5000 (CPU time: 32.375 s)
## MCMC scan 2700 of 5000 (CPU time: 33.406 s)
## MCMC scan 2800 of 5000 (CPU time: 34.609 s)
## MCMC scan 2900 of 5000 (CPU time: 35.938 s)
## MCMC scan 3000 of 5000 (CPU time: 37.250 s)
## MCMC scan 3100 of 5000 (CPU time: 38.672 s)
## MCMC scan 3200 of 5000 (CPU time: 39.859 s)
## MCMC scan 3300 of 5000 (CPU time: 41.000 s)
## MCMC scan 3400 of 5000 (CPU time: 42.328 s)
## MCMC scan 3500 of 5000 (CPU time: 43.469 s)
## MCMC scan 3600 of 5000 (CPU time: 44.750 s)
## MCMC scan 3700 of 5000 (CPU time: 46.031 s)
## MCMC scan 3800 of 5000 (CPU time: 47.344 s)
## MCMC scan 3900 of 5000 (CPU time: 48.672 s)
## MCMC scan 4000 of 5000 (CPU time: 49.969 s)
## MCMC scan 4100 of 5000 (CPU time: 51.125 s)
## MCMC scan 4200 of 5000 (CPU time: 52.266 s)
## MCMC scan 4300 of 5000 (CPU time: 52.922 s)
## MCMC scan 4400 of 5000 (CPU time: 53.938 s)
## MCMC scan 4500 of 5000 (CPU time: 54.828 s)
## MCMC scan 4600 of 5000 (CPU time: 55.734 s)
```

```
## MCMC scan 4700 of 5000 (CPU time: 56.797 s)
## MCMC scan 4800 of 5000 (CPU time: 57.781 s)
## MCMC scan 4900 of 5000 (CPU time: 58.766 s)
## MCMC scan 5000 of 5000 (CPU time: 59.938 s)
```

Plot results

```
y_grid <- seq(min(y), max(y), length.out = 100) # grid for evaluating densities

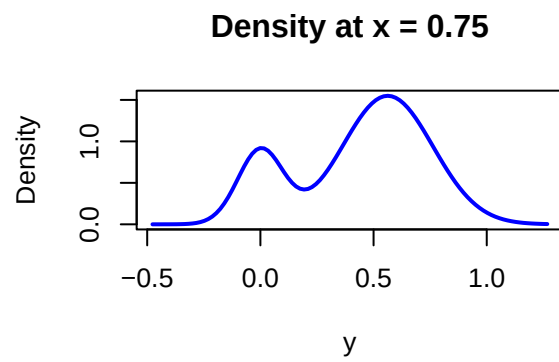
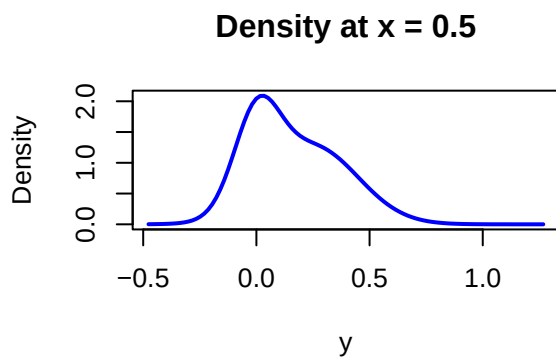
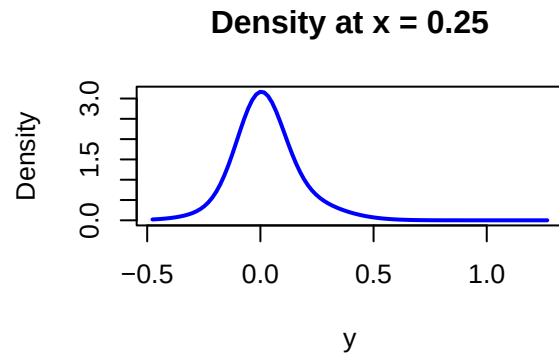
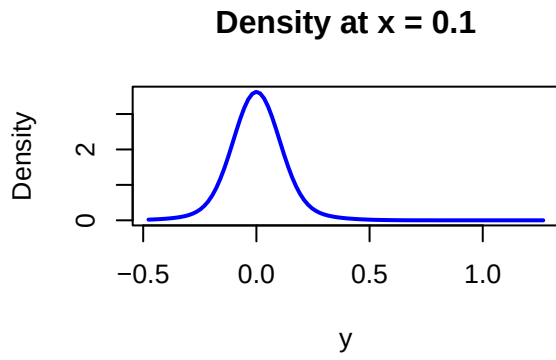
# compute conditional densities
x_values <- c(0.1, 0.25, 0.5, 0.75) # values of x to evaluate
density_results <- lapply(x_values, function(x_val) {
  # weights
  weights <- exp(-2 * x_val)

  # component densities
  y1_density <- dnorm(y_grid, mean = 0, sd = 0.1) # Component 1
  y2_density <- dnorm(y_grid, mean = x_val^2, sd = 0.2) # Component 2

  # combined density
  conditional_density <- weights * y1_density + (1 - weights) * y2_density

  list(grid = y_grid, density = conditional_density)
})

par(mfrow = c(2, 2))
for (i in seq_along(x_values)) {
  plot(
    density_results[[i]]$grid, density_results[[i]]$density,
    type = "l", col = "blue", lwd = 2,
    xlab = "y", ylab = "Density",
    main = paste("Density at x =", x_values[i])
  )
}
```



The plots show the transition of the conditional density. We can see that for $x = 0.1$ and 0.25 the density is unimodal but for 0.5 and 0.75 it becomes bimodal where for $x = 0.75$ it becomes more pronounced